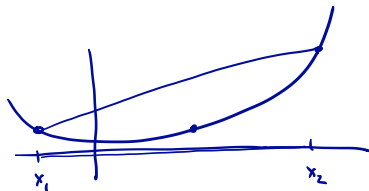


Convex functions - definition 1: no assumptions

$$\begin{array}{l} \min: f(x) \\ \text{st: } x \in \mathcal{X} \end{array}$$

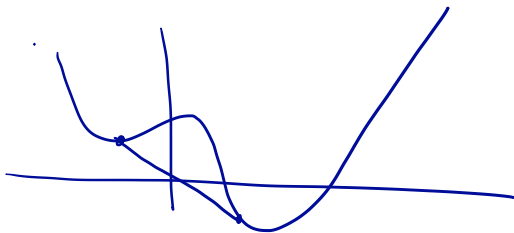


$f: \mathbb{R}^n \rightarrow \mathbb{R}$, f is convex if

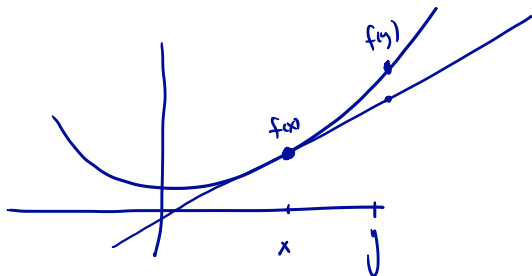
$$\forall x_1, x_2 \in \mathbb{R}^n, \quad \forall \lambda \in [0, 1]$$

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

Not convex:

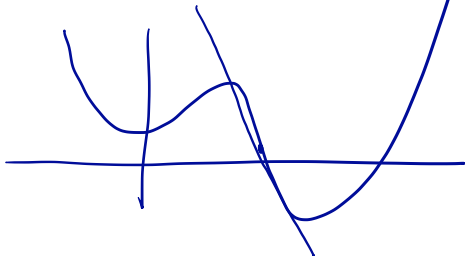


Convex functions - definition 2: gradient exists

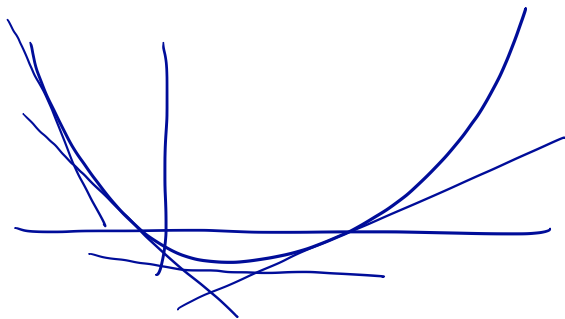


A fn f is convex if:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \forall y$$



Convex functions - definition 3: Hessian exists

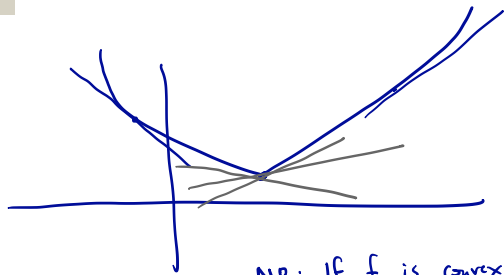


$$1-D: f''(x) \geq 0$$

A function f is convex if: $\nabla^2 f(x)$ is psd,
i.e., all e-values of $\nabla^2 f(x)$ are ≥ 0

$$\nabla^2 f(x) \succeq 0$$

Convex functions - definition 2' (non-smooth)



A function is convex if
 $\forall x, \exists g$ s.t.

$$f(y) \geq f(x) + \langle g, y-x \rangle$$

NB: If f is convex & differentiable, then $g = \nabla f(x)$
satisfies this - it is the only such g .

the subgradient and subdifferential

For a convex function f , a vector g s.t.

$$f(y) \geq f(x) + \langle g, y-x \rangle \quad \forall y, \text{ is called a } \underline{\text{sub-gradient}}$$

The set of subgradients of f at a point x , is called

the subdifferential of f at x : $\partial f(x)$

Applying the three definitions: $x^T Q x$

$$f(x) = x^T Q x, \quad Q \succeq 0$$

Definition 3:

$$\nabla^2 f(x) = 2Q \succeq 0 \quad \checkmark$$

Definition 2: $f(y) \stackrel{?}{\geq} f(x) + \langle \nabla f(x), y-x \rangle \quad \forall y$

$$\nabla f(x) = 2Qx$$

$$y^T Q y \stackrel{?}{\geq} x^T Q x + \langle 2Qx, y-x \rangle$$

$$y^T Q y \stackrel{?}{\geq} x^T Q x + 2x^T Q (y-x)$$

$$y^T Q y - 2x^T Q (y-x) - x^T Q x \stackrel{?}{\geq} 0$$

$$y^T Q y - 2x^T Q y + x^T Q x = (y-x)^T Q (y-x) \geq 0 \quad \checkmark$$

Applying the three definitions: $x'Qx$

Definition 1:

Exercise. Need to check:

$$\forall x, y, \quad \forall \lambda \in [0, 1] : f(\lambda x + (1-\lambda)y) \stackrel{?}{\leq} \lambda f(x) + (1-\lambda)f(y)$$

plug in and check that the inequality holds

hint: similar calculation as above.

More on definition of convex functions

f is convex :

$$1. f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

$$2. f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \forall y$$

$$f(y) \geq f(x) + \langle g, y - x \rangle \quad \forall y$$

$$3. \nabla^2 f(x) \succeq 0$$

The three definitions are equivalent (1/3)

Definition 3 implies definition 2 (and the fundamental theorem of calculus)

The subgradient/gradient of a convex function is monotone

If f is convex, then the subgradient (gradient)
is monotone:

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$$

$$\langle g_x - g_y, x - y \rangle \geq 0 \quad \text{for } g_x \in \partial f(x), g_y \in \partial f(y)$$

$$f(y) \geq f(x) + \langle g_x, y - x \rangle$$

$$f(x) \geq f(y) + \langle g_y, x - y \rangle$$

$$\cancel{f(x)} - \cancel{f(y)} \geq \cancel{f(x)} - \cancel{f(y)} + \langle g_x - g_y, y - x \rangle$$

$$\langle g_x - g_y, x - y \rangle \geq 0$$

The three definitions are equivalent (2/3)

$$f \text{ monotone} : \langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$$

If f satisfies definition 3, it is monotone

$$\text{If } \nabla^2 f(x) \succeq 0 : \quad (\text{Recall: } \int_0^1 F'(t) dt = F(1) - F(0))$$

$$\begin{aligned} \int_0^1 (x-y)^T \nabla^2 f(tx + (1-t)y) dt &= \int_0^1 \frac{d}{dt} \left(\nabla f(tx + (1-t)y) \right) dt \\ &= \nabla f(x) - \nabla f(y) \end{aligned}$$

Take inner prod with $(x-y)$:

$$0 \leq \underbrace{\int_0^1 (x-y)^T \nabla^2 f(tx + (1-t)y) (x-y) dt}_{u(\nabla^2 f)u} = \langle \nabla f(x) - \nabla f(y), x - y \rangle$$

$$\text{So: } \nabla^2 f(x) \succeq 0 \Rightarrow f \text{ monotone}$$

The three definitions are equivalent (3/3)

If f is monotone, it is convex according to def 2.

$$\begin{aligned}\int_0^1 \nabla f((y-x)t + x)^T (y-x) dt &= \int_0^1 \frac{d}{dt} f((y-x)t + x) dt \\ &= f(y) - f(x) \\ \Rightarrow f(y) &= f(x) + \underbrace{\int_0^1 \nabla f((y-x)t + x)^T (y-x) dt}_{\geq \langle \nabla f(x), y-x \rangle} \quad \begin{array}{l} \text{integrand eval'd at } t=0 \\ \text{monotone property} \end{array}\end{aligned}$$

We can show this inequality holds, by showing that the integrand is smallest at $t=0$. We will use monotone property.

$$\begin{aligned}\text{Monotone: } &\langle \nabla f((y-x)t + x) - \nabla f(x), (y-x)t + x - x \rangle \geq 0 \\ \Leftrightarrow &\underbrace{\langle \nabla f((y-x)t + x) - \nabla f(x), y-x \rangle}_{h(t) - h(0) \geq 0} \geq 0 \quad \int h(t) \geq h(0) \cdot 1\end{aligned}$$

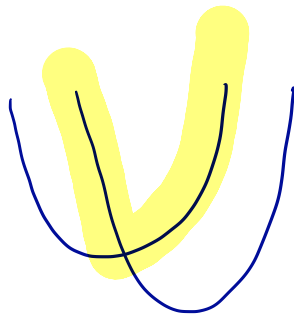
$$\Rightarrow f(y) = f(x) + \int_0^1 h(t) dt \geq f(x) + h(0) = f(x) + \langle \nabla f(x), y-x \rangle$$



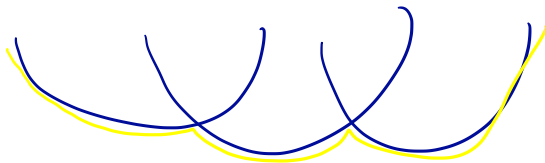
The Max of Convex Functions is Convex

Exercise: Translate this
into algebra using
def 1 of convexity.

proof by picture:



the min may not be:



Largest element of a vector

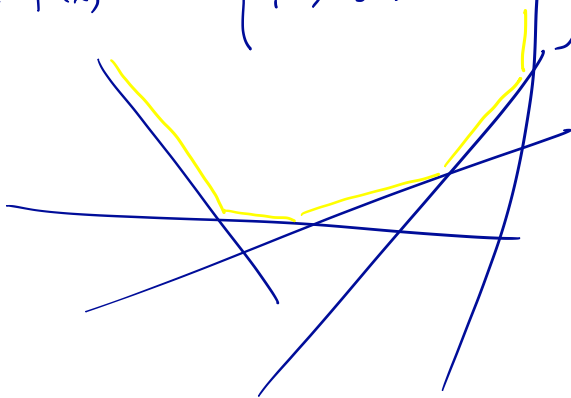
$f(x_1, \dots, x_n) = \text{maximum element}$ (defined on $\mathbb{R}_+^n$)

ex: $f(3, 0, 5) = 5$, $f(2, 1, \frac{1}{2}) = 2$

$$f(x_1, \dots, x_n) = f(x) = \max \{e_1^T x, e_2^T x, \dots, e_n^T x\}$$

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$f_i(x) = e_i^T x = x_i$$



Largest eigenvalue of a symmetric matrix

$$\underline{f(Q)} = \lambda_{\max}(Q)$$

To use previous fact, we need to show that

$\lambda_{\max}(Q)$ = max of convex functions.

Recall : Q symmetric $\Rightarrow x^T Q x \leq \lambda_{\max} \cdot \|x\|_2^2$

and $x^T Q x = \lambda_{\max} \cdot \|x\|_2^2$ when

x = eigenvector corresponding to $\lambda_{\max}$.

$$x^T Q x = x^T (\lambda_{\max} x) = \lambda_{\max} x^T x$$

$$= \lambda_{\max} \|x\|^2$$

$$\lambda_{\max} = \sup_{\substack{x^T Q x \\ \text{st. } \|x\|_2 = 1}}$$

$$\left. \begin{array}{l} x^T Q x \text{ as a function of } \underline{Q} \text{ is linear} \\ \underline{\langle x x^T, Q \rangle} = x^T Q x \end{array} \right\}$$